CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (IT)

November – December 2011

IT 202 Data Communication

Date: 30.11.2011, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 **Do as Direct.**

(a) Define Passive Circuit. **[01]**

(b) A 400 – Watt carrier is modulated to a depth of 75 percent. Calculate the total power in the modulated wave. **[02]**

(c) The Available S/N input to output power ratio of an amplifier is 7.94:1. Calculate the output signal to noise ratio when the input signal to noise ratio is 35 dB. **[03]**

(d) Define transformation ratio in tapped inductor. **[01]**

Q - 2 (a) Draw the block diagram of superhetrodyne receiver and show the output at various points. **[08]**

(b) Describe self capacitance of coil with its necessary mathematical equation. **[06]**

OR

Q - 2 (a) What is capacitive tap? Explain in detail. **[06]**

(b) Write a note on sensitivity and gain of receiver. **[08]**

Q - 3 (a) Write short note on thermal noise. **[04]**

(b) The antenna current of an AM transmitter is 8 amperes when only the carrier is sent, but it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percent of modulation changes to 0.8. **[05]**

(c) Write short note on RC Phase Shift oscillator. **[05]**

OR

Q - 3 (a) Write short note on shot noise. **[04]**

(b) A certain transmitter radiates 9 kW with the carrier unmodulated, and 10.125 kW when the carrier is sinusoidally modulated. Calculate the modulation index, percent of modulation. If another sine wave, corresponding to 40 percent modulation, is transmitted simultaneously, determine the total radiated power. **[05]**

(c) Write short note on Crystal oscillator. **[05]**

SECTION – II

Q – 4 Do as Direct

(a) Define Noise Factor. **[01]**

(b) List the different type of tracking error of superheterodyne receiver and which is best suitable? **[02]**

(c) Draw the frequency spectrum of an Amplitude Modulated wave and explain its component in detail. **[03]**

(d) Burst Noise is also called as a ________________. **[01]**

Q – 5 (a) Write short note on different properties of Fourier transform. **[06]**

(b) Draw the generalized hybrid model for BJT and derive its following parameter. **[08]**

1. current gain
2. input impedance
3. voltage gain
4. output impedance

OR

Q – 5 (a) Draw the hybrid model for three different configuration of bipolar junction transistor. **[08]**

(b) What is periodic and aperiodic signal? Compare power and energy signals. **[06]**

Q – 6 (a) List the different application of voltage controlled oscillator. **[02]**

(b) Give the difference between ideal filter and practical filter. **[02]**

(c) Compare series tuned circuit and parallel tuned circuit. **[03]**

(d) Explain the fundamental concept of DSBSC with its proper equation. **[03]**

(e) A mixer stage has a noise figure of 20 dB, and this is preceded by an amplifier that has a noise figure of 9 dB and an available power gain of 15 dB. Calculate the overall noise figure referred to the input. **[04]**

OR

Q – 6 (a) Describe energy spectral density with its properties **[02]**

(b) What is frequency synthesizer? Give its different type and also give the disadvantage of PLL frequency synthesizer **[02]**

(c) Which effect is occurring in thick conductor and high frequency application? Describe it. **[03]**

(d) Draw the block diagram of AM broadcast transmitter. **[03]**

(e) Derive the friis's formula for N number of amplifier connected in cascade connection. **[04]**

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